

Unit-4 Euler and Hamilton graphs and Graph colouring

Euler and Hamiltonian graphs and Graph Colouring: Definition of Euler Graphs, Hamiltonian Graphs, Standard Theorems on Euler and Hamiltonian graphs, planar graph, Dual of planar graphs, Graph coloring, Chromatic polynomial, Five Color Theorem, Matching, Cut set, Network flow, and its applications.

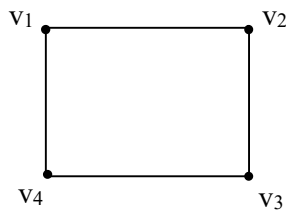
Euler Circuit and Euler Trial:

In a connected graph G if there is a circuit which contains all the edges of G then it is said to be Euler circuit in graph G .

In a connected graph G if there is a trial which contains all the edges of G then it is said to be Euler trial in graph G .

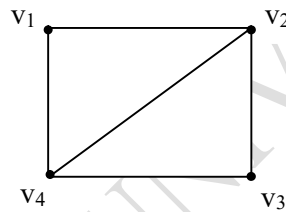
Example:

Euler Circuit



$V_1V_2V_3V_4V_1$

Euler trial



$V_2V_3V_4V_2V_1V_4$

Euler Graph: (Dec 2012)

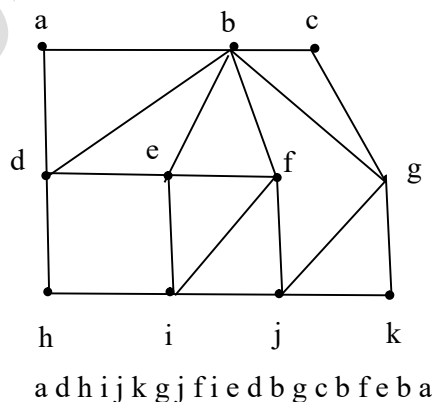
A connected graph that contains Euler circuit is called Euler graph.

Semi Euler Graph:

A connected graph that contains Euler trial is called Semi Euler graph.

Example: (June⁰⁶ 2010)

Find the Euler circuit for the graph



Theorem:

A connected graph G has an Euler circuit if and only if all the vertices of G are of even degree.

Proof:

Suppose G has an Euler circuit, while tracing the circuit we observe that every time the circuit meets a vertex v and goes through with two edges incident on v . This is true for all the vertices that belong to the circuit. Since the circuit contains all the edges it meets all the vertices at least once, hence the degree of every vertex is multiple of two.

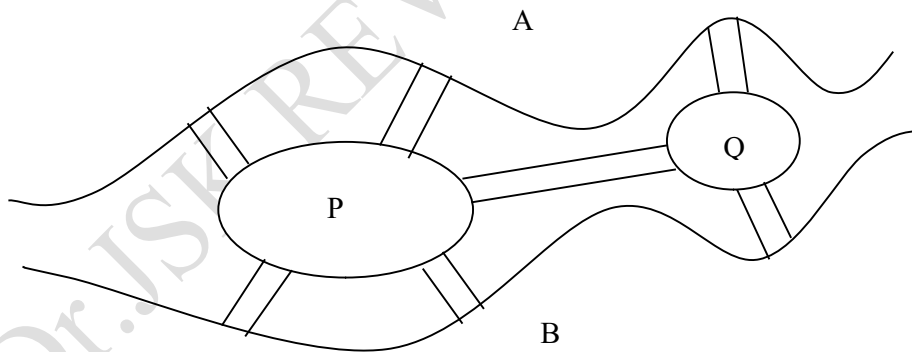
Conversely,

Suppose all the vertices of the graph G are of even degree. Now we construct a circuit starting at an arbitrary vertex v and going through the edges of G such that no edge is traced more than once. Since every vertex is of even degree we depart from every vertex we enter and tracing cannot be stopped at any vertex other than v . In this way we can obtain a circuit having v as starting and ending vertex which contain all the edges of the graph G and this circuit is called as Euler circuit of G .

The Konigsberg Bridge Problem:

(Discuss Konigsberg bridge problem and the solution of the problem)

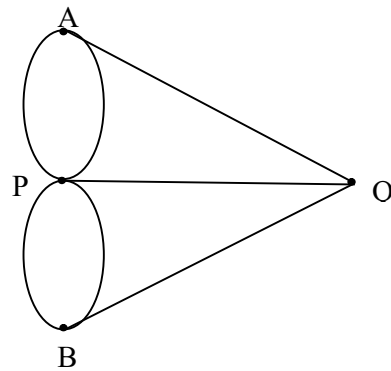
There are two islands and two land areas with seven bridges between them as shown in the figure



The problem is to start at any four areas of the city A, B, P, Q walk over any of the seven bridges exactly once and return to the starting point.

Solution:

Construct a graph by denoting the four land areas as four vertices and the seven bridges connecting them as seven edges



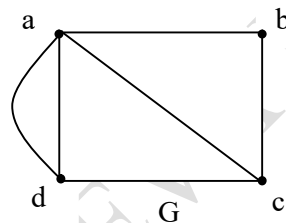
When we have to walk through all the edges and come back to the original vertex i.e., we find an Eulerian circuit in a graph. However in this graph, the degree of each vertex is not even ($\deg A = \deg B = \deg O = 3$, $\deg P = 5$) meaning that the Euler circuit does not exist in the graph. G is not an Euler graph. Hence there is no solution to this problem.

Hamilton Cycle and Hamilton Path:

Hamilton Cycle:

In a connected graph G , if there is a cycle that contains all the vertices of the graph, then it is called as Hamilton Cycle of the graph.

Example:



abcd is the Hamilton cycle of the graph.

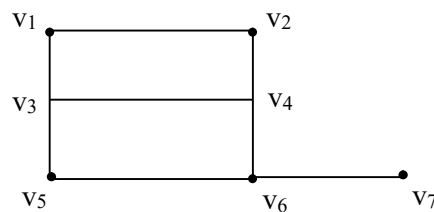
Hamilton Graph:

A Graph that contains a Hamilton cycle is called a Hamilton Graph.

Hamilton Path:

A path of a connected graph G is said to be the Hamilton path if it contains all the vertices of the graph G .

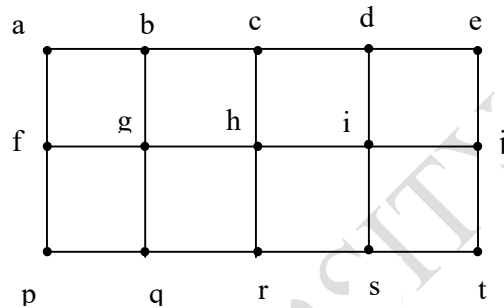
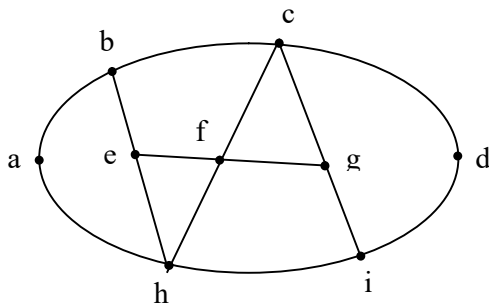
Example:



$v_1v_2v_4v_3v_5v_6v_7$ is the Hamilton path of G .

Example: (June⁰⁶ 2012)

Find a Hamilton cycle, if it exists, for each of the graphs shown in the following figures. If graph has no Hamilton cycle, determine whether it has a Hamilton path.

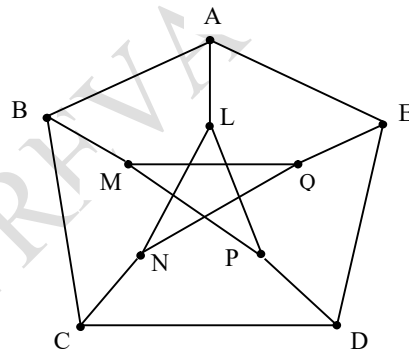


Example: (Jan⁰⁶ 2009)

Show that Peterson graph has no Hamilton cycle but it has a Hamilton path.

Solution:

The Peterson graph is a 3-regular graph with 10 vertices and 15 edges. The graph is shown below with the vertices labelled as A, B, C, D, E, L, M, N, P, Q.



Since the graph has 10 vertices and 15 edges, a Hamilton cycle if any in the graph must pass through all the vertices and must have 15 edges.

We observe that three edges are incident at every vertex of the graph. Of these three edges only two can be included in a Hamilton cycle (if it exists). Thus, at each of the 10 vertices of the graph one edge has to be excluded. By actual counting, we find that the number of edges to be excluded is 6. (For example, the edges AL, BM, CN, DE, PM, QE). Consequently, the number of edges that remain in the graph is $15 - 6 = 9$. These edges are insufficient to form a Hamilton cycle in the graph.

Thus, the Peterson graph does not contain a Hamilton cycle in it.

By examining the figure, we note that the edges AB, BC, CD, DE, EQ, QM, MP, PL, LN form a path and this path includes all the vertices. This path is therefore a Hamilton path.

Thus, the Peterson graph does have a Hamilton path in it.

Example: (Dec 2012)

Define Hamiltonian and Eulerian graphs. Prove the complete graph $K_{3,3}$ is Hamiltonian but not Eulerian.

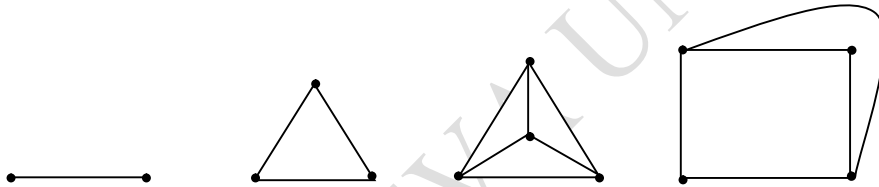
Example: (July 2014)

Define Hamilton Cycle. If $G = (V, E)$ is a loop free undirected graph with $|V| = n \geq 3$ and if $|E| = \left(\frac{n-1}{2}\right) + 2$, then prove that G has a Hamilton cycle.

Planar Graph:

A graph which can be drawn in a plane in which the edges meet only at the vertices is called a Planar Graph. Otherwise it is called non-planar graph.

Examples:



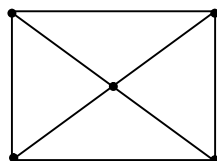
These are planar graphs because edges of these graphs meet only at vertices.

Examples: Show that i) a graph of order 5 and size 8.

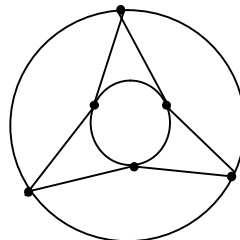
ii) a graph of order 6 and size 12 are planar.

Solution:

i)



(5,8) graph



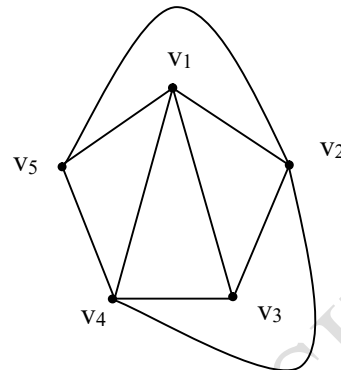
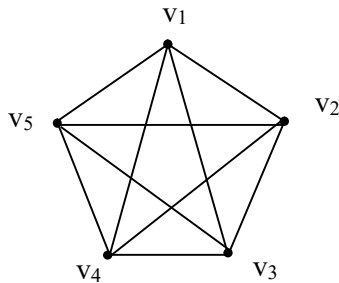
(6, 12) graph

These two graphs are planar graphs because the edges are meeting at vertices only.

Example:

Show that K_5 graph is non planar graph.

Solution:

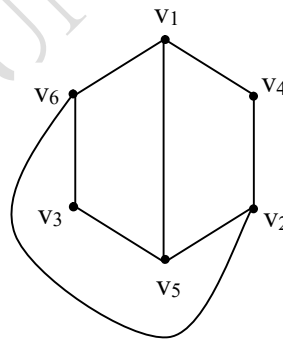
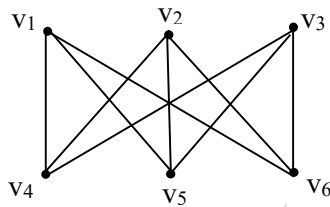


K_5 is not a planar graph because the edge between v_3 and v_5 in the second graph is not possible to draw without crossing the edges.

Example: (Dec⁰⁶ 2010, Dec⁰⁶ 2011)

Show that $K_{3,3}$ graph is a non planar graph.

Solution:

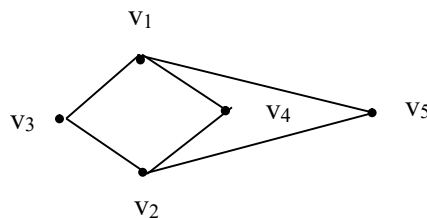
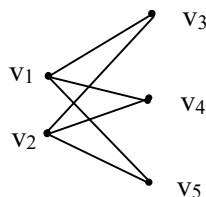


$K_{3,3}$ is not a planar graph because it is not possible to draw an edge between v_3 and v_4 either inside or outside the hexagon without crossing any edges.

Example:

Show that $K_{2,3}$ graph is a planar graph.

Solution:



$K_{2,3}$ is a planar graph because it can be drawn in a plane such that edges meet only at the vertices of the graph.

Euler's Fundamental Theorem:

Theorem: (Dec⁰⁶ 2010, July⁰⁶ 2011, July 2013, July 2014)

A connected planar graph G with n vertices and m edges has exactly $m - n + 2$ regions in all of its diagrams.

(If a connected planar graph G has n vertices, e edges and r regions, then prove that $n - e + r = 2$)(July 2013, July 2014)

Proof:

Consider a graph G with n vertices and m edges. Let r denote the number of regions in the diagram of G then the theorem states that

$$r = m - n + 2 \quad (\text{or}) \quad n - m + r = 2 \quad (1)$$

We prove this theorem by Mathematical induction on m .

If $m = 0$, then n must be equal to 1 i.e. $n = 1$, because G is a connected planar graph. Therefore if $n = 1$ then the diagram of G determines only one region of G that is the entire plane region as shown in figure (a)



Fig. (a)

Thus if $m = 0$, $n = 1$, $r = 1$, then (1) verifies as

$$r = m - n + 2$$

$$1 = 0 - 1 + 2 \text{ i.e., } 1 = 1$$

This shows that the theorem is true for $m = 0$.

Now assume that the theorem is true for all graph with $m = k$ number of edges where k is a non negative integer.

Consider a graph G_{k+1} with $k + 1$ edges and n vertices. Suppose that G_{k+1} has no cycles in it, then the diagram of G_{k+1} is in the form of figure (b).

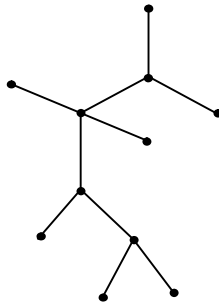


Fig. (b)

Therefore in this graph the number of vertices are exactly one more than the number of edges and the diagram will determine only one region. Hence in this case $n = (k + 1) + 1$ and $r = 1$.

Therefore (1) becomes

$$r = m - n + 2$$

$$1 = (k + 1) - (k + 1) - 1 + 2$$

$$1 = 1$$

This shows that (1) is true when $m = k + 1$ with no cycle in it.

Suppose G_{k+1} contains at least one cycle and r is the number of regions in the diagram G_{k+1} .

Consider an edge 'e' in a cycle and remove it from G_{k+1} , then the resulting graph is $G_{k+1} - e$. this will contain n vertices and $k + 1 - 1 = k$ edges. And the diagram determines $r - 1$ regions.

Since $G_{k+1} - e$ contains k edges the theorem holds for this graph by assumption.

$$\text{i.e., } r - 1 = k - n + 2$$

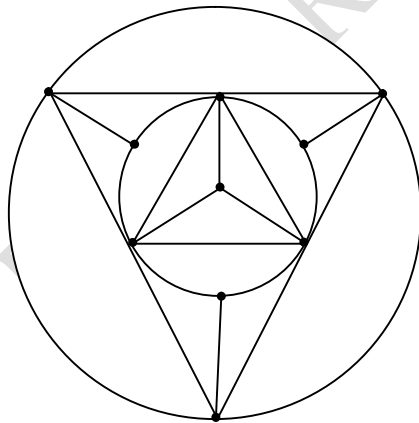
$$\text{Thus } r = (k + 1) - n + 2$$

This shows that in this case also the theorem is true when $m = k + 1$.

Hence by induction it follows that the result is true for all non negative integer m .

Example:

Find the number of vertices, edges and regions for the following planar graph, and verify the Euler's Theorem for these graphs. Fig (a)



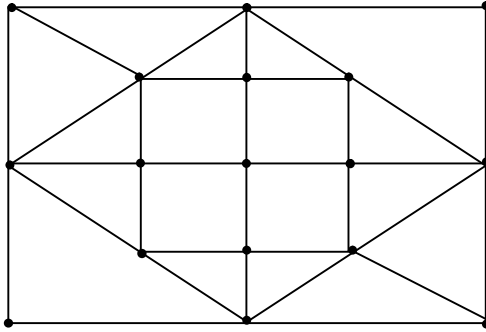
In Figure (a) $n = 10, m = 24, r = 16$

$$r = m - n + 2$$

$$16 = 24 - 10 + 2$$

$$16 = 16$$

Fig. (b) (July⁰⁶ 2008)



In Figure (b) $n = 17$, $m = 34$, $r = 19$

$$r = m - n + 2$$

$$19 = 34 - 17 + 2$$

$$19 = 19$$

Theorem: (June 2008, Jan⁰⁶ 2009, July⁰⁶ 2009, Jan⁰⁶ 2010)

Statement:

Prove that if G is connected simple planar graph with n (≥ 2) edges and r regions then

$$\text{i) } m \geq (3/2)r \quad \text{ii) } m \leq 3n - 6$$

Further if G is triangle free then, iii) $m \leq 2n - 4$.

Proof:

Since the graph G is planar graph, therefore it will not contain multiple edges and loops. Therefore every region must be bounded by 3 or more edges. Hence the total number of edges that bounds the regions is $\geq 3r$. On the other hand an edge is a boundary of at most 2 regions, therefore the total number of edges that bound all the regions is $\leq 2m$.

Thus $3r \leq 2m$

$$2m \geq 3r$$

$$m \geq (3/2)r \quad (1)$$

By substituting $r = m - n + 2$ in (1) we get

$$2m \geq 3(m - n + 2)$$

$$2m \geq 3m - 3n + 6$$

$$3n - 6 \geq 3m - 2m$$

$$3n - 6 \geq m$$

$$\Rightarrow m \leq 3n - 6 \quad (2)$$

If G is a triangle free graph then every region is bounded by at least 4 edges.

Therefore the total number of edges that bound all the regions is $\geq 4r$

Hence $4r \leq 2m$

Substituting $r = m - n + 2$

$$4(m - n + 2) \leq 2m \Rightarrow 4m - 4n + 8 \leq 2m$$

$$2m \leq 4n - 8$$

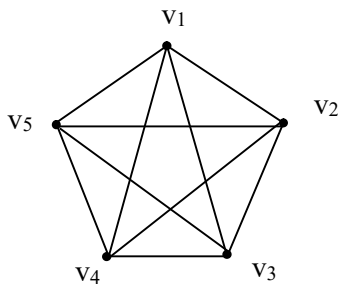
$$m \leq 2n - 4 \quad (3)$$

Example: (Jan⁰⁶ 2009)

Show that K_5 is a non-planar graph.

Solution:

K_5 have $n = 5$ vertices and $m = 10$ edges



For a connected simple planar graph we have

$$m \leq 3n - 6 \Rightarrow 10 \leq (3 \times 5) - 6$$

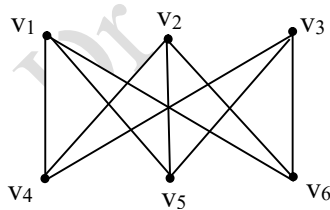
$$10 \leq 9$$

This is not true hence K_5 is non planar.

Example: (Jun 2008, Jan⁰⁶ 2009, Dec 2010)

Show that $K_{3,3}$ is a non-planar graph.

Solution:



$K_{3,3}$ have $n = 6$ vertices and $m = 9$ edges

Since $K_{3,3}$ is triangle free graph

For a connected simple triangle free planar graph we have

$$m \leq 2n - 4 \Rightarrow 9 \leq (2 \times 6) - 4$$

$9 \leq 8$ This is not true hence $K_{3,3}$ is non planar.

Example: (July⁰⁶ 2010)

Show that when any edge is removed from K_5 , the resulting subgraph is planar. Is this true for the graph $K_{3,3}$.

Example:

Every connected simple planar graph G contains a vertex of degree less than 6.

Solution:

Suppose every vertex of G is of degree greater than or equal to 6. Then, if $d_1, d_2, d_3, \dots, d_n$ are of degree of the n vertices of G , we have $d_1 \geq 6, d_2 \geq 6, d_3 \geq 6, \dots, d_n \geq 6$. Adding these, we get

$$d_1 + d_2 + d_3 + \dots + d_n \geq 6n$$

By handshaking property, the left hand side of this inequality is equal to $2m$, where m is the number of edges in G .

$$\text{Thus, } 2m \geq 6n \text{ or } 3n \leq m.$$

$$\text{On the other hand, } m \leq 3n - 6.$$

Thus, $3n \leq 3n - 6$. This cannot be true. Therefore, G must have a vertex of degree less than 6.

Example: (June⁰⁶ 2011, July 2012)

Show that every connected simple planar graph with less than 12 vertices must have a vertex of degree ≤ 4 .

Solution:

Suppose every vertex of G has degree greater than 4. Then, if $d_1, d_2, d_3, \dots, d_n$ are the degrees of n vertices of G , we have $d_1 \geq 5, d_2 \geq 5, d_3 \geq 5, \dots, d_n \geq 5$, so that

$$d_1 + d_2 + d_3 + \dots + d_n \geq 5n \text{ or } 2m \geq 5n, \text{ by hand shaking property}$$

$$\text{or } (5n/2) \leq m \tag{1}$$

$$\text{But we know that } m \leq 3n - 6$$

From (1) we get

$$(5n/2) \leq 3n - 6$$

$$\Rightarrow n \geq 12 \tag{2}$$

Thus, if every vertex of G has degree greater than 4, then G must have at least 12 vertices. Hence, if G has less than 12 vertices, it must have a vertex of degree ≤ 4 .

Dual of a Graph:

Let $G = (V, E)$ be a planar graph. To construct a dual of G place a point inside each region determined by G . For each edge shared by two regions draw an edge connecting the vertices inside these regions. For an edge lying entirely in one region draw a loop intersecting the edge exactly once.

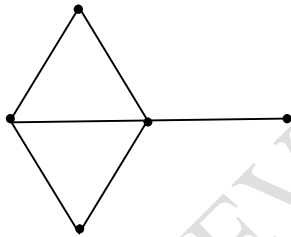
The Dual of G is denoted by G^* or G^d .

Observations:

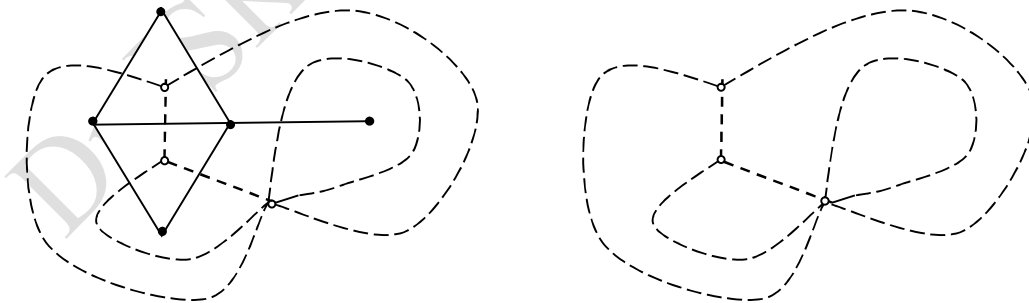
- 1) There is one to one correspondence between regions of G and the vertices of G^* .
- 2) There is one to one correspondence between edges of G and the edges of G^* .
- 3) Corresponding to a loop in G there is a pendent vertex in G^* .
- 4) Corresponding to a pendent vertex in G there is a loop in G^* .
- 5) Like G , G^* is also connected planar graph.

Example: (Feb 2006)

Find the dual of the graph G . Write down any four observations of G and its dual.

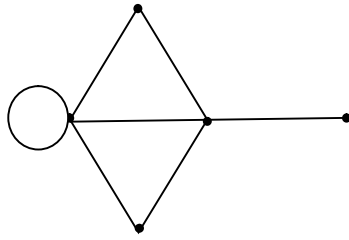


Solution:

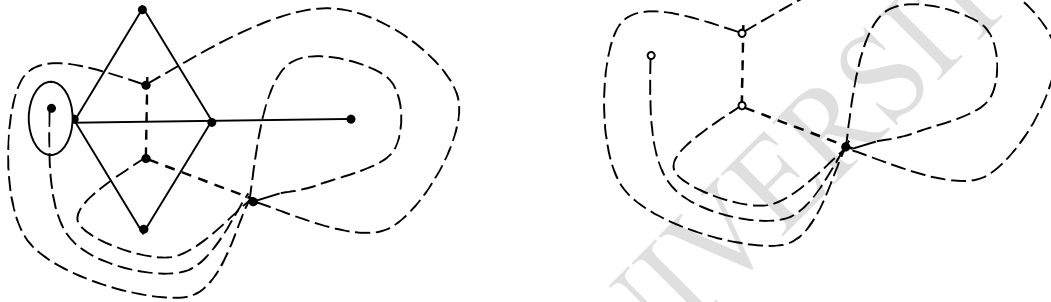


Example:

Find the dual of the graph

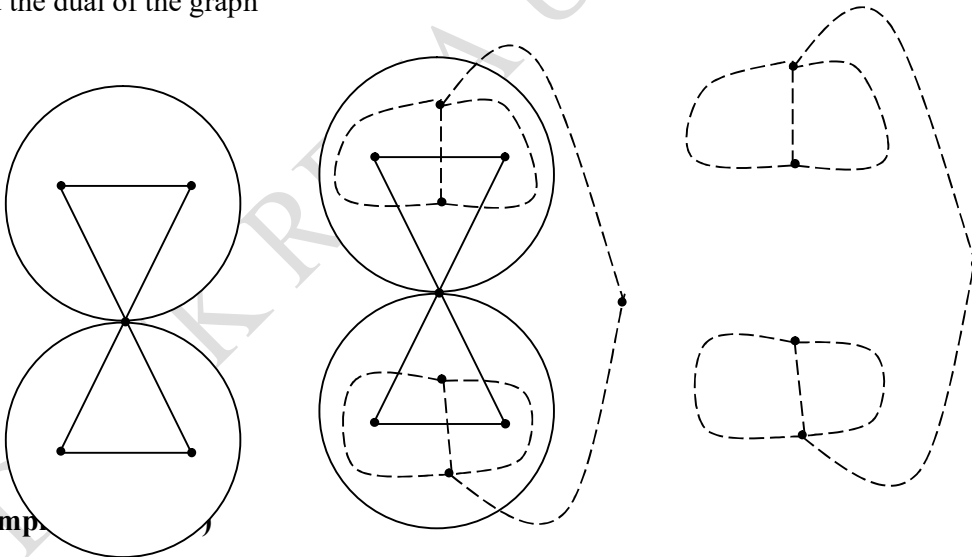


Solution:



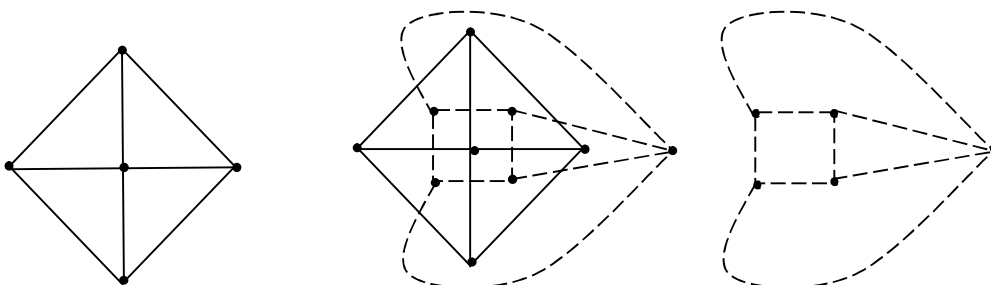
Example: (Jan⁰⁶ 2009)

Find the dual of the graph



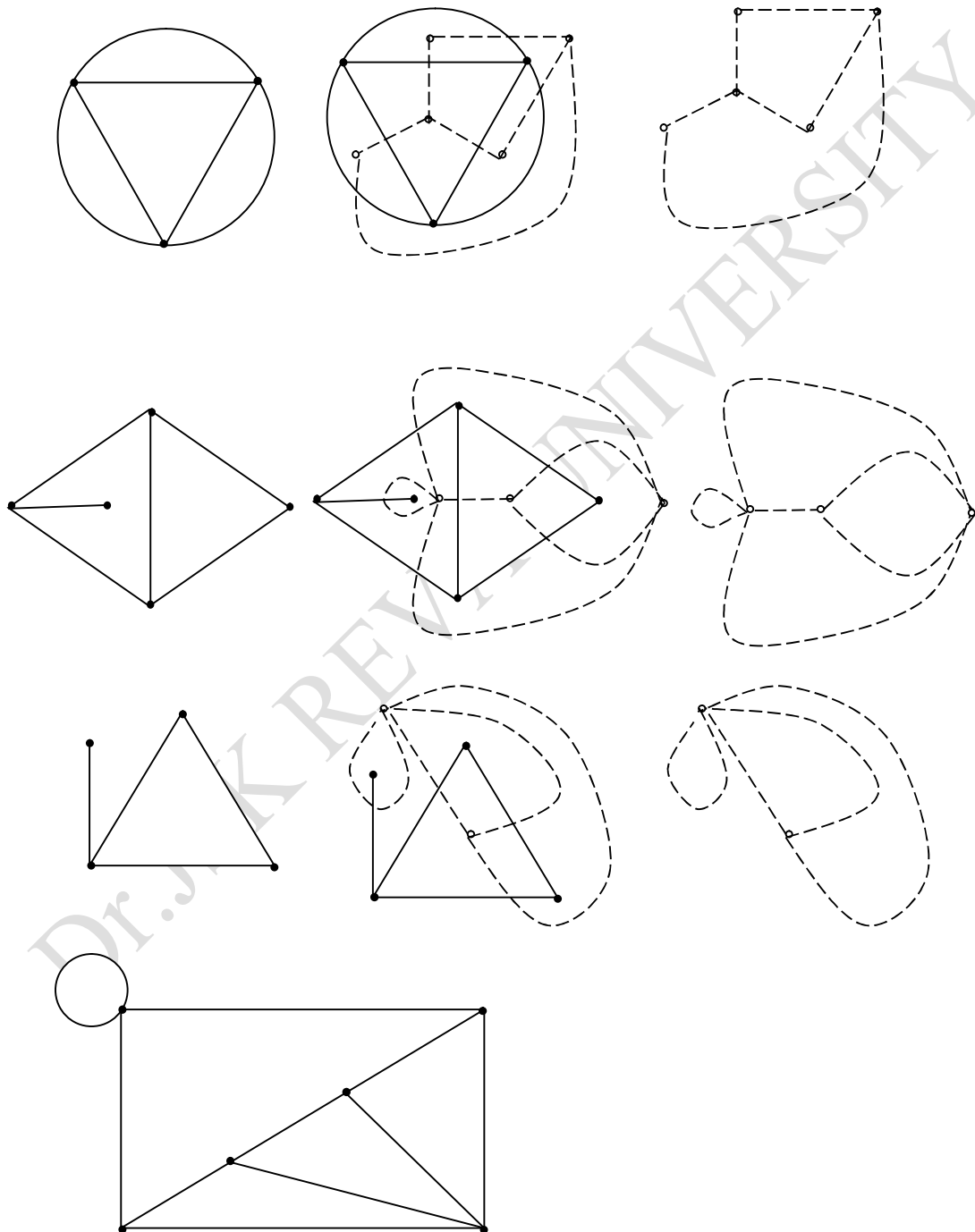
Example:

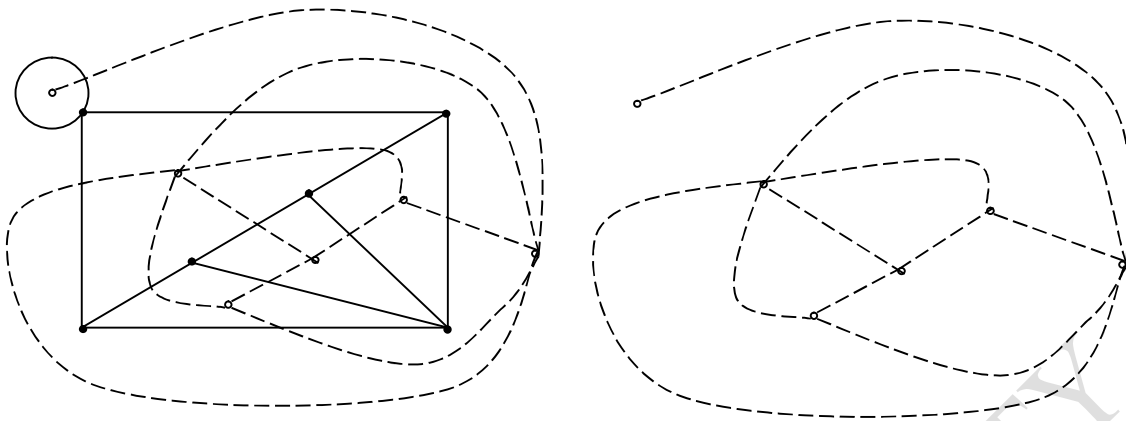
Find the dual of the graph



Example: (Aug 2004)

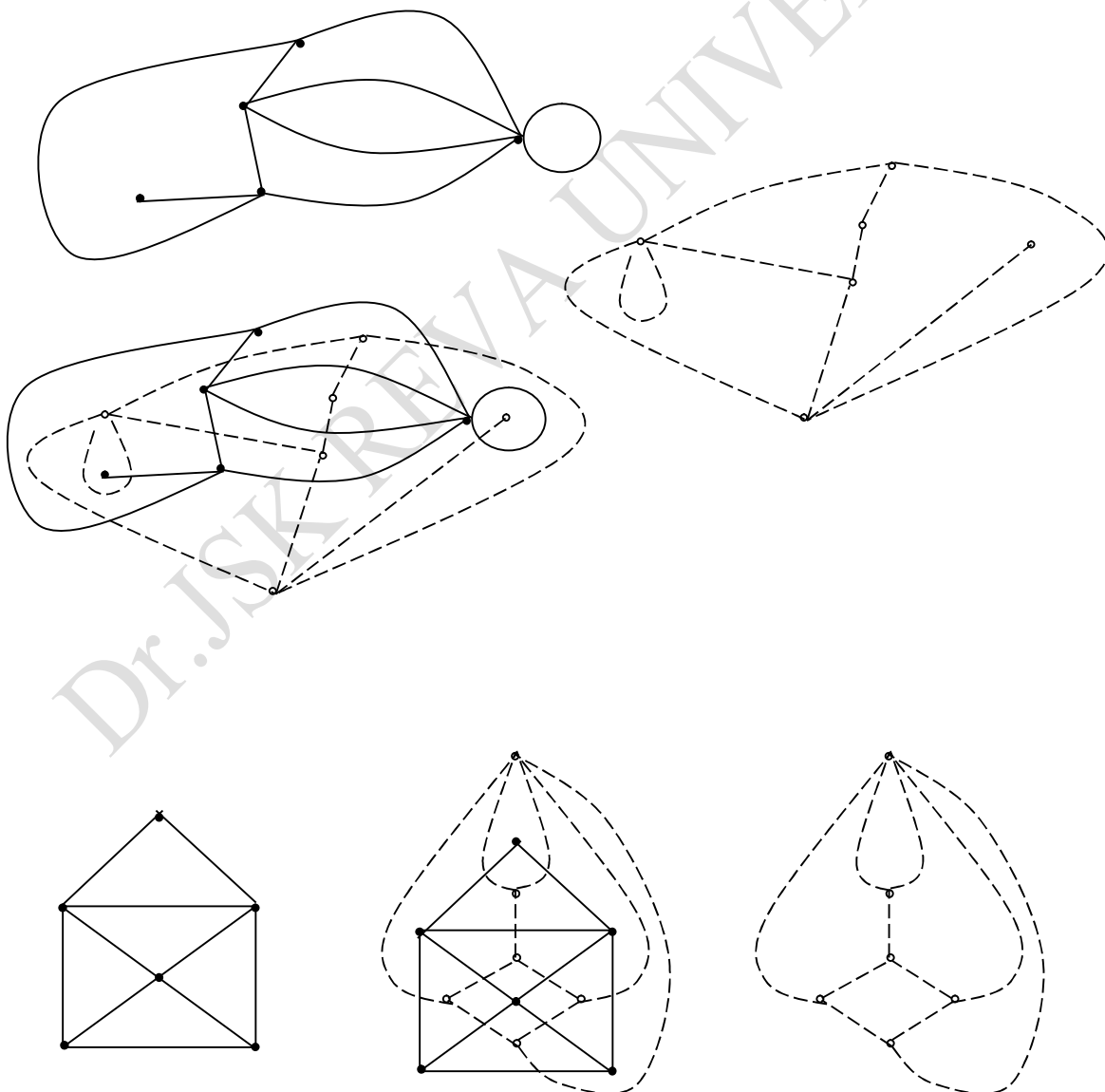
Find the dual of the graph





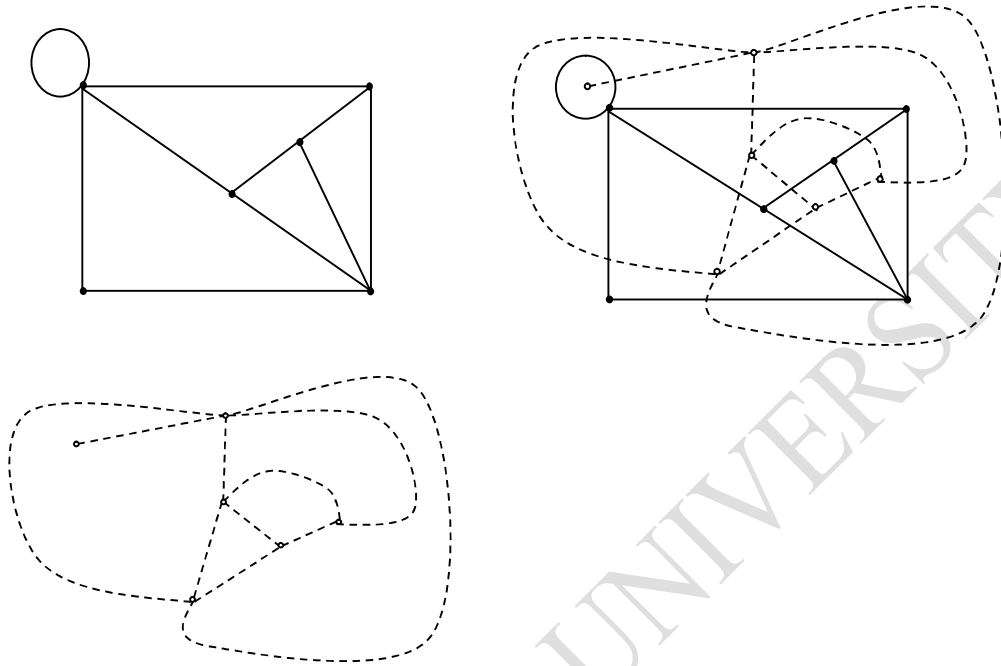
Example: (Jan 2014)

Define dual of a planar graph. Draw the geometric dual of the given graph.



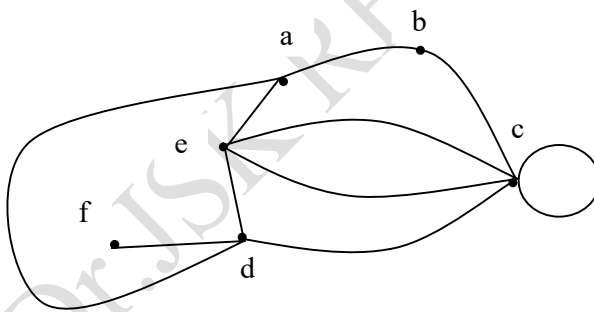
Example: (Dec 2014)

Construct the dual of the planar graph given below.



Example: (Dec⁰⁶ 2011)

Find the dual of the following graph



Graph Colouring:

Graph colouring is colouring the vertices of a Graph.

Properly Colouring:

It is colouring of the graph with no two adjacent vertices coloured with same colour.

Chromatic Number:

If a graph G is properly coloured with minimum number of colours, then that number is called Chromatic number of the graph G .

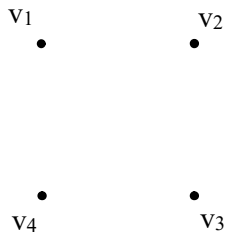
(OR)

Chromatic number is the minimum number of colours required to properly colour a graph G .

It is denoted by $\chi(G)$.

Example:

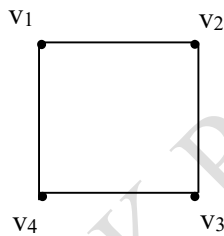
1)



The given graph is 1 – Chromatic

or $\chi(G) = 1$

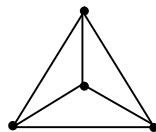
2)



The given graph is 2 – Chromatic

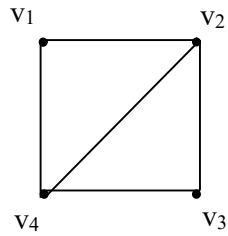
or $\chi(G) = 2$

3)



The given graph is 4 – Chromatic or $\chi(G) = 4$

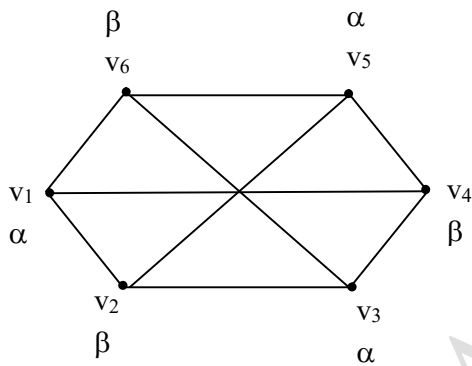
4)



The given graph is 3 – Chromatic or $\chi(G) = 3$.

Example:

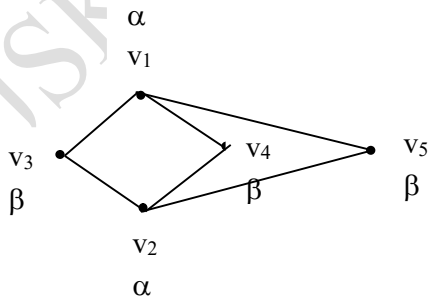
Find the Chromatic number of the graph



We can assign one colour say α for the vertices v_1, v_3, v_5 which are not mutually adjacent and to the vertices v_2, v_4, v_6 we can assign the colour β which is different from α .

Therefore $\chi(G) = 2$

2)



$\chi(G) = 2$

Chromatic Polynomial:

Let G be an undirected graph and λ be the number of colours available for properly colouring the vertices of G . The chromatic polynomial is the polynomial function in the variable λ that gives the number of different ways in which we can colour the vertices of G using all λ colours.

The chromatic polynomial for the graph G is denoted by $P(G, \lambda)$.

Chromatic Polynomial for Different Graphs:

1) If the given graph contains n number of vertices and no edges i.e., null graph with n vertices, then the chromatic polynomial is defined as

$$P(N_n, \lambda) = \lambda^n.$$

2) If the given graph is complete graph with n vertices then the chromatic polynomial is defined as

$$(a) P(K_n, \lambda) = 0 \text{ if } \lambda < n$$

$$(b) P(K_n, \lambda) = n! \text{ if } \lambda = n$$

$$(c) P(K_n, \lambda) = \lambda(\lambda - 1)(\lambda - 2) \dots \{\lambda - (n-1)\} \\ = \lambda(\lambda - 1)(\lambda - 2) \dots (\lambda - n + 1) \text{ if } \lambda > n$$

3) If the graph contains a path with n vertices then the chromatic polynomial is defined as

$$P(L_n, \lambda) = \lambda(\lambda - 1)^{n-1}$$

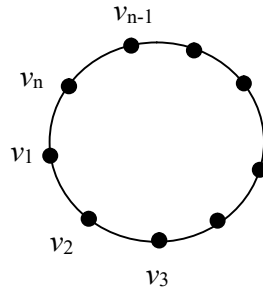
Here λ must be ≥ 2 .

Example: (July 2013)

Find the chromatic number of the complete bipartite graph $K_{m,n}$ and a cycle C_n on n vertices, $n \geq 3$.

Solution:

The number of colours needed does not depend upon m and n . However, only two colours are needed to colour the set of m vertices with one colour and the set of n vertices with second colour. Since, edges connect only a vertex from the set of m vertices and a vertex from the set of n vertices, no two adjacent vertices have the same colour.



The cycle being considered is shown in the above figure. Obviously, this cycle cannot be properly coloured with single colour. Assign two colours alternately to the vertices, starting with v_1 . Then, the odd vertices v_1, v_3, v_5 etc., will have one colour α and the even vertices v_2, v_4, v_6 etc., will have a different colour β .

Suppose n is even. Then the vertex v_n is an even vertex and therefore will have the colour β , and the cycle gets properly coloured (because the adjacent vertices v_n and v_1 have different colours). Therefore, the cycle is 2 – chromatic.

Suppose n is odd. Then the vertex v_n is an odd vertex and therefore will have the colour α , and the cycle is not properly coloured (because the adjacent vertices v_n and v_1 have the same colours α). Thus, in this case, the cycle is 3 – chromatic.

Example:

Define chromatic number. Prove that a graph of order n (≥ 2) consisting of a single circuit is 2 – chromatic if n is even, and 3 – chromatic if n is odd.

Example: (Dec 2012)

Prove that a graph is 2 – chromatic if and only if it is a null bipartite graph.

Solution:

Suppose a graph G is 2 – chromatic. Then it is non-null, and some vertices of G have one colour, say α , and rest of the vertices have another colour, say β . Let V_1 be the set of vertices having colour α and V_2 be the set of vertices having colour β . Then $V_1 \cup V_2 = V$, the vertex set of G , and $V_1 \cap V_2 = \phi$. Also, no two vertices of V_1 can be adjacent and no other vertices of V_2 can be adjacent. As such, every edge in G has one end in V_1 and the other end in V_2 . Hence G is bipartite graph.

Conversely, suppose G is a non-null bipartite graph. Then the vertex set of G has two bipartites V_1 and V_2 such that every edge in G has one end in V_1 and another end in V_2 . Consequently, G cannot be properly coloured with one colour, because then vertices in V_1 and V_2 will have the same colour and every edge has both of its ends of the same colour. Suppose we assign a colour α to all vertices in V_1 and a different colour β to all vertices in V_2 . This will make a proper colouring of V . Hence G is 2-chromatic.

Product Rule:

Let G be a graph with n components $G_1, G_2, G_3, \dots, G_n$ then

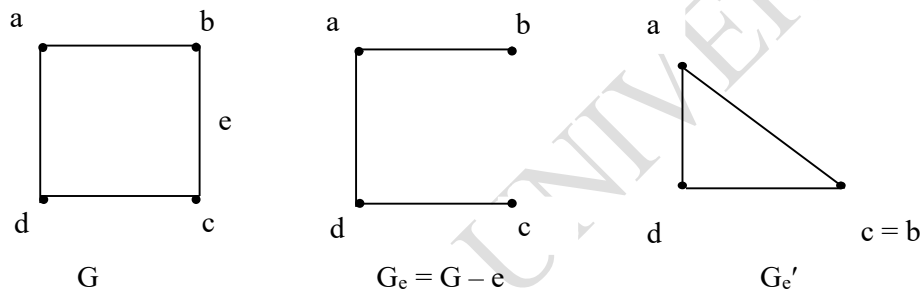
$$P(G, \lambda) = P(G_1, \lambda) P(G_2, \lambda) P(G_3, \lambda) P(G_4, \lambda) \dots P(G_n, \lambda)$$

is called as the product rule.

Decomposition Theorem: (June⁰⁶ 2010 only statement)**Statement:**

If G is a connected graph and e is an edge in the graph G then

$$P(G_e, \lambda) = P(G, \lambda) + P(G_{e'}, \lambda)$$

Proof:

In a proper colouring of graph G_e the vertices b and c can take same or different colours.

In every proper colouring of G the vertices b and c can take different colours.

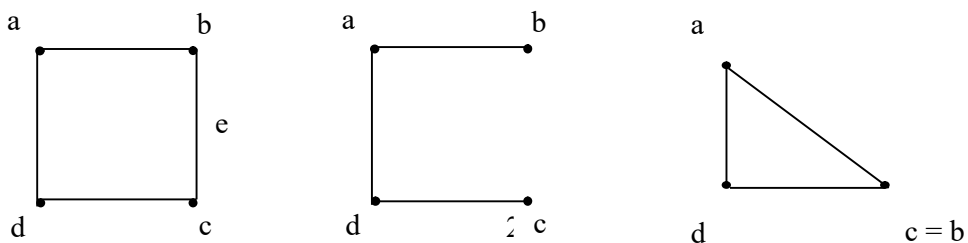
In every proper colouring of $G_{e'}$ these vertices will take the same colour.

Therefore the number of proper colouring of G_e is the sum of number of proper colouring of graph G and number of proper colouring of graph $G_{e'}$.

$$\text{i.e., } P(G_e, \lambda) = P(G, \lambda) + P(G_{e'}, \lambda)$$

$$(\text{or}) P(G, \lambda) = P(G_e, \lambda) - P(G_{e'}, \lambda).$$

Example: Find the chromatic polynomial of the cycle C_4 of length 4. Hence find its chromatic number.

Solution:

G

$G_e = G - e$

G_e'

By decomposition theorem we have

$$P(G_e, \lambda) = P(G, \lambda) + P(G_e', \lambda)$$

$$P(G, \lambda) = P(G_e, \lambda) - P(G_e', \lambda)$$

$$P(G, \lambda) = \lambda(\lambda - 1)^{4-1} - \lambda(\lambda - 1)(\lambda - 2) \quad | \quad P(K_n, \lambda) = \lambda(\lambda - 1)(\lambda - 2) \dots (\lambda - n + 1)$$

$$= \lambda(\lambda - 1)^3 - \lambda(\lambda - 1)(\lambda - 2)$$

$$= \lambda(\lambda - 1) \{ (\lambda - 1)^2 - (\lambda - 2) \} = (\lambda^2 - \lambda) \{ \lambda^2 - 2\lambda + 1 - \lambda + 2 \}$$

$$= (\lambda^2 - \lambda) \{ \lambda^2 - 3\lambda + 3 \} = \lambda^4 - 3\lambda^3 + 3\lambda^2 - \lambda^3 + 3\lambda^2 - 3\lambda$$

$$P(G, \lambda) = \lambda^4 - 4\lambda^3 + 6\lambda^2 - 3\lambda$$

The given graph is 2 – Chromatic or $\chi(G) = 2$

Example: (Feb 2006)

Find the chromatic polynomial of the graph given below. What is its chromatic number.



Solution:

The given graph is a path

$$\text{Therefore } P(L_n, \lambda) = \lambda(\lambda - 1)^{n-1}$$

$$P(L_5, \lambda) = \lambda(\lambda - 1)^{5-1}$$

$$P(L_5, \lambda) = \lambda(\lambda - 1)^4 \text{ and } \chi(G) = 2.$$

Multiplication Theorem:

Statement:

If G is a graph containing sub graphs G_1 and G_2 such that $G = G_1 \cup G_2$ and $G_1 \cap G_2 = K_n$ then

$$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda)}{\lambda^{(n)}}$$

where $\lambda^{(n)} = \lambda(\lambda - 1)(\lambda - 2) \dots (\lambda - n + 1)$.

Proof:

Since $K_n = G_1 \cap G_2$ and as we know that $\psi(K_n) = n$. It follows that K_n is sub graph of G_1 and G_2 and $\chi(G_1) \geq n$ and $\chi(G_2) \geq n$

Given $\lambda > n$ number of different colours, there are $\lambda^{(n)} = \lambda(\lambda - 1)(\lambda - 2) \dots (\lambda - n + 1)$ number of proper colourings of K_n . For each of these $\lambda^{(n)}$ proper colourings of K_n , the product rule gives $P(G_1, \lambda) / \lambda^{(n)}$ ways of properly colouring the remaining vertices of G_1 . Similarly, there are $P(G_2, \lambda) / \lambda^{(n)}$ ways of properly colouring the remaining vertices of G_2 . As such,

$$P(G, \lambda) = P(K_n, \lambda) \frac{P(G_1, \lambda)}{\lambda^{(n)}} \frac{P(G_2, \lambda)}{\lambda^{(n)}}$$

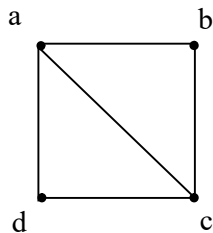
$$P(G, \lambda) = \lambda^{(n)} \frac{P(G_1, \lambda)}{\lambda^{(n)}} \frac{P(G_2, \lambda)}{\lambda^{(n)}}$$

$$P(G, \lambda) = \frac{P(G_1, \lambda) P(G_2, \lambda)}{\lambda^{(n)}}$$

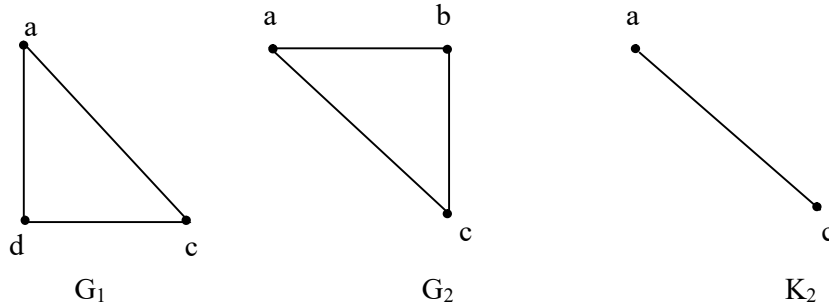
This completes the proof of the theorem.

Example:

Find the chromatic polynomial and chromatic number of the graph. If four colours are used in how many ways can the graph be properly coloured



Solution:



The given graph G is regarded as the union of G_1 and G_2 shown in figure and $G_1 \cap G_2 = K_2$, also G_1 and G_2 are same as K_3 .

By multiplication theorem

$$P(G, \lambda) = \frac{P(G_1, \lambda)P(G_2, \lambda)}{\lambda^{(n)}}$$

$$P(G, \lambda) = \frac{P(K_3, \lambda)P(K_3, \lambda)}{\lambda^{(2)}}$$

$$P(G, \lambda) = \frac{\lambda(\lambda-1)(\lambda-2) \times \lambda(\lambda-1)(\lambda-2)}{\lambda(\lambda-1)}$$

$$P(G, \lambda) = \lambda(\lambda-1)(\lambda-2)^2$$

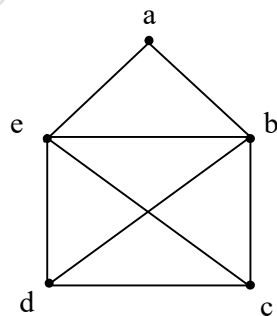
When $\lambda = 4$

$$P(G, 4) = 4(4-1)(4-2)^2 = 32$$

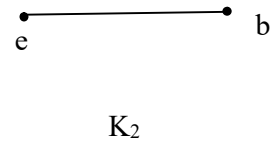
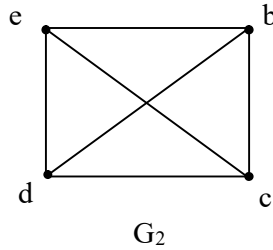
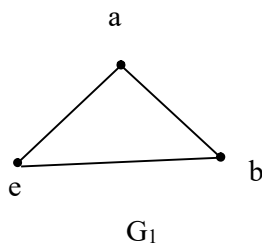
If four colours are used then 32 ways the graph can be properly coloured.

Example:

Find the chromatic polynomial of the graph.



Solution:



The given graph G is regarded as the union of G_1 and G_2 shown in figure and $G_1 \cap G_2 = K_2$, also G_1 and G_2 are same as K_3 .

By multiplication theorem

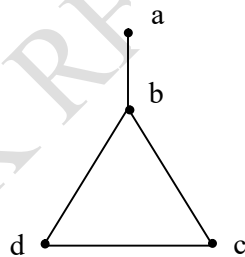
$$P(G, \lambda) = \frac{P(G_1, \lambda)P(G_2, \lambda)}{\lambda^{(n)}}$$

$$P(G, \lambda) = \frac{P(K_3, \lambda)P(K_4, \lambda)}{\lambda^{(2)}}$$

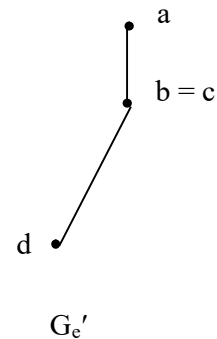
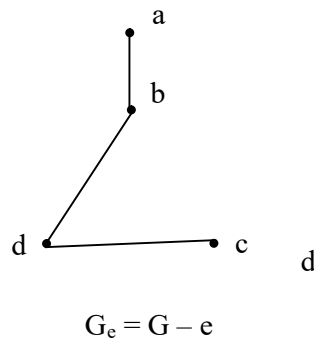
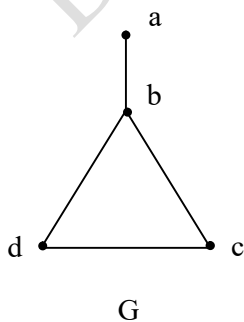
$$P(G, \lambda) = \frac{\lambda(\lambda-1)(\lambda-2) \times \lambda(\lambda-1)(\lambda-2)(\lambda-3)}{\lambda(\lambda-1)} = \lambda(\lambda-1)(\lambda-2)^2(\lambda-3)$$

Example: (June 2008, Jan2014)

Find the chromatic polynomial of the graph. Hence find the colouring of graph with 4 colours.



Solution:



By decomposition theorem we have

$$P(G_e, \lambda) = P(G, \lambda) + P(G_{e'}, \lambda)$$

$$P(G, \lambda) = P(G_e, \lambda) - P(G_{e'}, \lambda)$$

$$\begin{aligned} P(G, \lambda) &= \lambda(\lambda - 1)^{4-1} - \lambda(\lambda - 1)^{3-1} & | P(L_n, \lambda) &= \lambda(\lambda - 1)^{n-1} \\ &= \lambda(\lambda - 1)^3 - \lambda(\lambda - 1)^2 \\ &= \lambda(\lambda - 1)^2 \{ (\lambda - 1) - 1 \} \\ &= \lambda(\lambda - 1)^2(\lambda - 2) \end{aligned}$$

$$P(G, \lambda) = \lambda(\lambda - 1)^2(\lambda - 2)$$

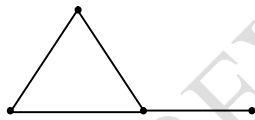
When $\lambda = 4$

$$P(G, 4) = 4(4-1)^2(4-2) = 72$$

If four colours are used then 72 ways the graph can be properly coloured

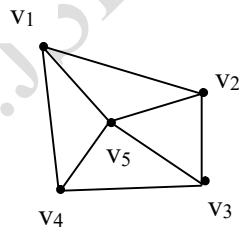
Example: (Dec 2014)

Determine the chromatic polynomial for the graph G in the following figure



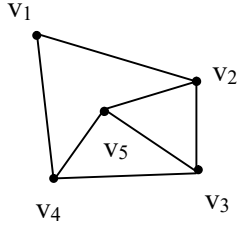
Example: (Jan⁰⁶ 2009, Dec⁰⁶ 2010, June⁰⁶ 2012, June 2015)

Find the chromatic polynomial for the graph given below

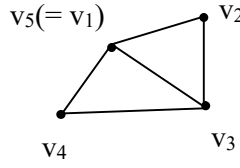


Solution:

Let us denote the given graph by G and the edge $\{v_1, v_5\}$ as e. Then the graphs G_e and G'_e would be as shown below.

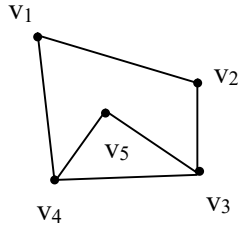


G_e

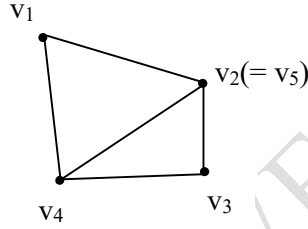


G'_e

Let us redesignate G_e as H and denote the edge $\{v_5, v_2\}$ as f . Then the graphs H_f and H'_f are as shown below.



H_f



H'_f

Now, H'_f is the union of the cycles $v_1v_4v_2v_1$ and $v_2v_3v_4v_2$ each of which is the same as K_3 , and that the intersection of these cycles is the edge $\{v_4, v_2\}$ which is the same as K_2 . Therefore, by the multiplication theorem, we have

$$P(H'_f, \lambda) = \frac{P(K_3, \lambda)P(K_3, \lambda)}{\lambda^{(2)}}$$

Similarly,

$$P(G'_e, \lambda) = \frac{P(K_3, \lambda)P(K_3, \lambda)}{\lambda^{(2)}}$$

Next, we note that H_f is the union of the cycles $v_1v_2v_3v_4v_1$ and $v_5v_3v_4v_5$ and that the intersection of these cycles is the edge $\{v_4, v_3\}$. The first of these cycles is C_4 , the second cycle is K_3 and the edge $\{v_4, v_3\}$ is K_2 . Therefore, by the multiplication theorem, we have

$$P(H_f, \lambda) = \frac{P(C_4, \lambda)P(K_3, \lambda)}{\lambda^{(2)}}$$

Now, by using the decomposition theorem and the fact that $H \equiv G_e$, we get

$$P(G, \lambda) = P(G_e, \lambda) - P(G'_e, \lambda)$$

$$P(G, \lambda) = P(H, \lambda) - P(G'_e, \lambda)$$

$$P(G, \lambda) = P(H_f, \lambda) - P(H'_f, \lambda) - P(G'_e, \lambda)$$

$$P(G, \lambda) = \frac{P(C_4, \lambda)P(K_3, \lambda)}{\lambda^{(2)}} - \frac{P(K_3, \lambda)P(K_3, \lambda)}{\lambda^{(2)}} - \frac{P(K_3, \lambda)P(K_3, \lambda)}{\lambda^{(2)}}$$

$$P(G, \lambda) = \frac{1}{\lambda^{(2)}} \{P(C_4, \lambda)P(K_3, \lambda) - 2P(K_3, \lambda)P(K_3, \lambda)\} = \frac{P(K_3, \lambda)}{\lambda^{(2)}} \{P(C_4, \lambda) - 2P(K_3, \lambda)\}$$

$$P(G, \lambda) = \frac{\lambda(\lambda-1)(\lambda-2)}{\lambda(\lambda-1)} \{ \lambda \{ (\lambda-1)^3 - (\lambda-1)(\lambda-2) \} - 2\lambda(\lambda-1)(\lambda-2) \}$$

$$| \text{ From previous example in page 22, } P(C_4, \lambda) = \lambda \{ (\lambda-1)^3 - (\lambda-1)(\lambda-2) \}$$

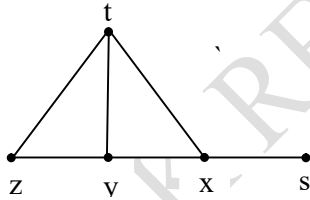
$$P(G, \lambda) = \frac{\lambda(\lambda-1)(\lambda-2)}{\lambda(\lambda-1)} \{ \lambda(\lambda-1)^3 - 3\lambda(\lambda-1)(\lambda-2) \}$$

$$P(G, \lambda) = \lambda(\lambda-1)(\lambda-2) \{ (\lambda-1)^2 - 3(\lambda-2) \}$$

$$P(G, \lambda) = \lambda(\lambda-1)(\lambda-2)(\lambda^2 - 5\lambda + 7)$$

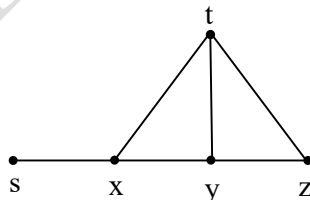
Example:

Determine the chromatic polynomial for the graph G in the following figure



Example:

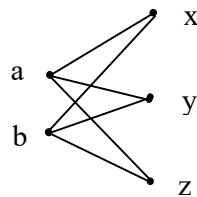
Determine the chromatic polynomial for the graph G in the following figure



Example:

Consider the graph $K_{2,3}$ shown in figure, and let $\lambda \in \mathbb{Z}^+$ denote the number of colours available to properly colour the vertices of $K_{2,3}$

- 1) How many proper colourings of $K_{2,3}$ have vertices a, b coloured the same.
- 2) How many proper colourings of $K_{2,3}$ have vertices a, b coloured with different colours.
- 3) What is the chromatic polynomial of $K_{2,3}$? What is $\chi(K_{2,3})$?

**Solution:**

1) If the vertices a and b are to have the same colour, then there are λ choices for colouring the vertex a and only one choice for the vertex b (or vice versa). Consequently, there are $\lambda - 1$ choices for each of the vertices x, y, z. Hence, the number of proper colourings (in this case) is $\lambda(\lambda - 1)^3$.

2) If the vertices a and b are to have different colours, then there are λ choices for colouring the vertex a and $(\lambda - 1)$ choices for the vertex b (or vice versa). Consequently, there are $\lambda - 2$ choices for each of the vertices x, y, z. Hence, the number of proper colourings (in this case) is $\lambda(\lambda - 1)(\lambda - 2)^3$.

iii) Since the two cases of the vertices a and b having the same colour or different colours are exhaustive and mutually exclusive, the chromatic polynomial of the graph is

$$P(K_{2,3}, \lambda) = \lambda(\lambda - 1)^3 + \lambda(\lambda - 1)(\lambda - 2)^3$$

$$\chi(K_{2,3}) = 2.$$

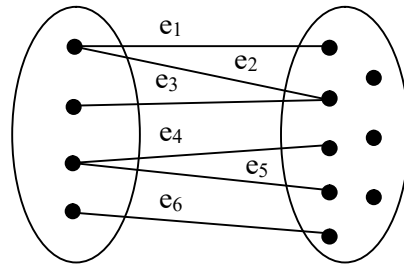
Matchings:

Consider a bipartite graph $G(V_1, V_2; E)$, where V_1 and V_2 are the two partitions of the vertex set of G and E is the edge set of G . A subset M of E is called a matching in G if no two edges that belong to M share a common vertex in V_1 and V_2 .

Complete Matching: (Dec⁰⁶ 2010, June⁰⁶ 2012, July 2014)

A matching M in G is called a complete matching from V_1 to V_2 if every vertex in V_1 is an end vertex of some edge that belongs to M .

Example:



$\{e_1, e_4, e_6\}$ = matching

$\{e_1, e_3, e_4, e_6\}$ = not a matching (e_4 and e_6 share common vertex in V_1)

$\{e_2, e_3, e_5, e_6\}$ = not a matching (e_4 and e_6 share common vertex in V_2)

$\{e_1, e_3, e_4, e_6\}$ = complete matching

$\{e_1, e_2, e_3, e_5, e_6\}$ = not a matching.

Consequences of the definition of a complete matching:

1) In a complete matching from V_1 to V_2 , corresponding to every vertex in V_1 there must exist a unique vertex in V_2 such that there is an edge between the two vertices. But, corresponding to a vertex in V_2 there may be a vertex in V_1 to which it is joined by an edge.

2) It is necessary that the number of vertices in V_2 is greater than or equal to the number of vertices in V_1 .

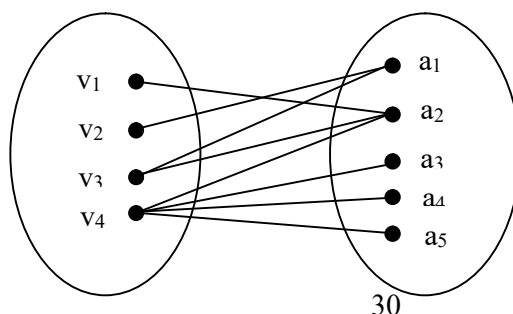
Hall's Theorem: (Provides a necessary and sufficient condition for the existence of a complete matching)

Statement:

In a bipartite graph $G = G(V_1, V_2; E)$, a complete matching from V_1 to V_2 exists if and only if every subset of k vertices in V_1 is collectively adjacent to k or more vertices in $V_2 \forall$ values of k .

Example:

Prove that the bipartite graph shown below does not have a complete matching from V_1 to V_2



V_1

V_2

Solution:

We observe that the three vertices v_1, v_2, v_3 in V_1 are together joined to two vertices a_1, a_2 in V_2 . Thus, there is a subset of 3 vertices in V_1 which is collectively adjacent to 2 (< 3) vertices in V_2 . Hence by Hall's theorem there does not exist complete matching from V_1 to V_2 .

Theorem: (June 2015)

Prove that in a bipartite graph $G(V_1, V_2, E)$ if there is a positive integer M such that the degree of every vertex in $V_1 \geq M \geq$ the degree of every vertex in V_2 , then there exists a complete matching from V_1 to V_2 .

Proof:

Consider any subset W of k vertices in V_1 . If the degree of every vertex in V_1 is greater than or equal to M for some positive integer M (as per the condition given), then there are at least M edges incident on each vertex in V_1 . Therefore, a total of at least Mk edges are incident on k vertices in W . Each of these Mk edges is incident on some vertex V_2 . Thus, there are at least Mk edges from W to V_2 . Since the degree of every vertex in V_2 is at most M (as per the condition given), at most M edges go to any vertex in V_2 . Therefore, if there are k' vertices in V_2 that are the end vertices of edges from W , then the number of edges from W to V_2 is at most Mk' . Accordingly, the number of edges from W to V_2 is at least Mk and at most Mk' . As such

$Mk \leq Mk'$, or $k' \geq k$.

Thus, under the given condition, any subset W of k vertices in V_1 is collectively adjacent to $k' \geq k$ vertices in V_2 . Therefore, by Hall's Theorem there exists a complete matching from V_1 to V_2 .

Example: (Dec 2012, July 2013)

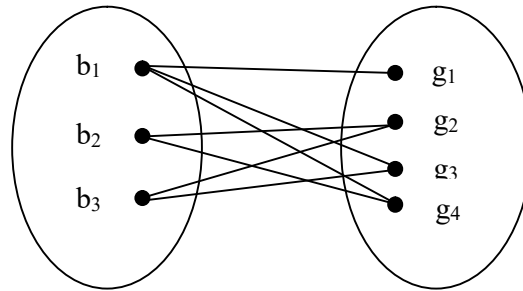
Three boys b_1, b_2, b_3 and four girls g_1, g_2, g_3, g_4 are such that

- (i) b_1 is a cousin of g_1, g_3, g_4
- (ii) b_2 is a cousin of g_2 , and g_4
- (ii) b_3 is a cousin of g_2 , and g_3

Can every one of the boys marry a girl who is one of his cousins? If so, find five possible sets of such couples. (Marriage problem)

Solution:

Let us draw a bipartite graph $G(V_1, V_2; E)$ in which V_1 consists of b_1, b_2, b_3 and V_2 consists of g_1, g_2, g_3, g_4 and E consists of edges representing the given cousin relationship.



Consider

k (No. of elements)	S_i (Subset of V_1)	S_i' (Adjacent vertices subset in V_2)
$k = 1$	$\{b_1\}$	$\{g_1, g_2, g_3\}$
	$\{b_2\}$	$\{g_1, g_2, g_4\}$
	$\{b_3\}$	$\{g_2, g_3, g_4\}$
$k = 2$	$\{b_1, b_2\}$	$\{g_1, g_2, g_3, g_4\}$
	$\{b_1, b_3\}$	$\{g_1, g_2, g_3, g_4\}$
	$\{b_2, b_3\}$	$\{g_2, g_3, g_4\}$
$k = 3$	$\{b_1, b_2, b_3\}$	$\{g_1, g_2, g_3, g_4\}$

We observe that for each S_i , the number of elements in S_i' is greater than or equal to the number of elements in S_i . Therefore, the graph has a complete matching by Hall's theorem. Thus each boy can marry a girl who is one of his cousins.

Five possible couple-sets are

Set 1: $\{b_1, g_1\}, \{b_2, g_2\}, \{b_3, g_3\}$.

Set 2: $\{b_1, g_1\}, \{b_2, g_4\}, \{b_3, g_2\}$.

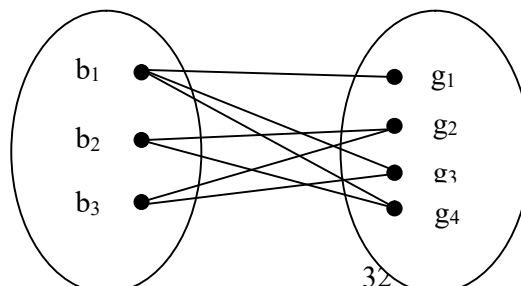
Set 3: $\{b_1, g_4\}, \{b_2, g_2\}, \{b_3, g_3\}$.

Set 4: $\{b_1, g_3\}, \{b_2, g_4\}, \{b_3, g_1\}$.

Set 5: $\{b_1, g_1\}, \{b_2, g_4\}, \{b_3, g_3\}$.

Example: (June 2013)

Show that the following graph has a complete matching from V_1 to V_2 . Obtain two complete matching



V_1

V_2

By examining the graph, we note that:

Degree of every vertex in $V_1 \geq 2 \geq$ degree of every vertex in V_2 .

Therefore, there is a complete matching from V_1 to V_2 .

Two complete matching from V_1 to V_2 are

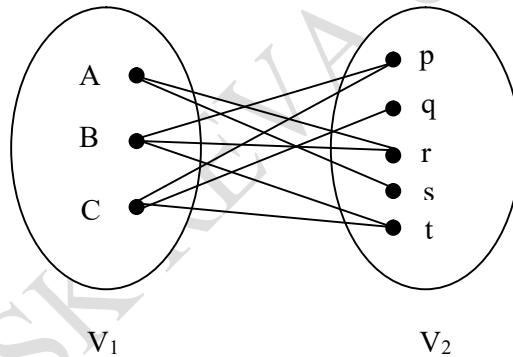
$\{b_1, g_1\}, \{b_2, g_2\}, \{b_3, g_3\}, \{b_1, g_1\}, \{b_2, g_4\}, \{b_3, g_3\}$

Example: (June⁰⁶ 2008)

Students p, q, r, s, t are members of three committees A, B, and C: r and s belong to committee A; p, r, t belong to committee B, and p, q and t belong to committee C. Each committee is to select a student representative. Can a selection be made in such a way that each committee has distinct representatives?

Solution:

The committee memberships of the five students can be depicted in a bipartite graph as shown below.



By examining the graph, we note that:

Degree of every vertex in $V_1 \geq 2 \geq$ degree of every vertex in V_2 .

Therefore, there is a complete matching from V_1 to V_2 .

This means that the three committees can select three different representatives.

For example, s can selected from A, r can selected from B and t can selected from C.

Disconnecting Edge Set:

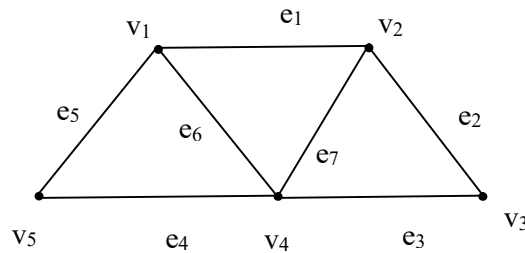
Let G be a connected graph. A set S of edges of G is said to be a disconnecting edge set in G if the removal of S from G disconnects G .

Cut Set:

A set S of edges of G is said to be a cut set in G if the following conditions hold

- i) S is a disconnected edge set in G .
- ii) No proper sub set of S is a disconnecting edge set in G .

Example:



$S_1 = \{e_4, e_5\}$ is a disconnecting edge set which is a cut set.

$S_2 = \{e_4, e_5, e_6\}$ is not a cut set because the proper sub set $\{e_4, e_5\}$ of S_2 is a disconnecting edge set.

Example:

Cut Sets:

$$S_1 = \{e_1, e_5, e_7\}$$

$$S_2 = \{e_1, e_2, e_6\}$$

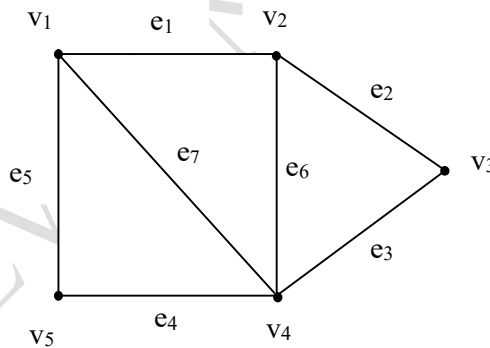
$$S_3 = \{e_2, e_3\}$$

$$S_4 = \{e_3, e_4, e_6, e_7\}$$

$$S_5 = \{e_4, e_5\}$$

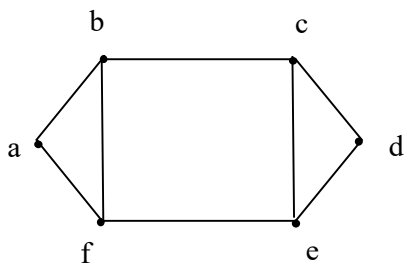
$$S_6 = \{e_2, e_5, e_6, e_7\}$$

$$S_7 = \{e_2, e_4, e_6, e_7\} \text{ and so on.}$$



Cut set with respect to a pair of vertices:

Let G be a connected graph and A and B be a pair of vertices of G . Then a cut set in G whose removal from G puts A and B in to two different parts of G is called a cut set w. r. t. A and B .



Cut sets w. r. t. the vertices a and d are

$$S_1 = \{ab, af\}, S_2 = \{bc, fe\}$$

$$S_3 = \{cd, ed\}, S_4 = \{ab, bf, fe\}$$

$$S_5 = \{ab, bf, ce, ed\}, S_6 = \{af, fb, bc\}$$

$$S_7 = \{af, fb, ce, cd\}, S_8 = \{bc, ce, ed\}$$

$$S_9 = \{fe, ec, cd\}$$

Edge connectivity:

Let G be a connected graph. The edge connectivity of G is the minimum number of edges of G whose removal results in a disconnected or trivial graph.

Vertex connectivity:

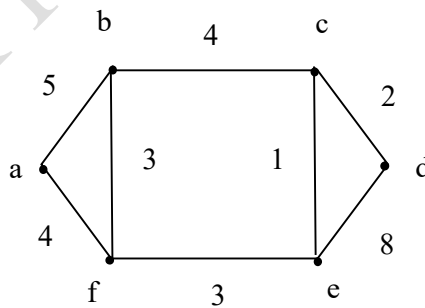
Let G be a connected graph. The minimum number of vertices whose removal results in a disconnected or trivial graph is called the vertex – connectivity.

Capacity of a cut-set:

Let G be a weighted connected graph and S be a cut-set in G . Then the sum of the weights of all edges in S is called the capacity of the cut-set S .

Example:

For the network shown in figure, find the capacities of all the cut-sets between the vertices a and d and hence determine the maximum flow between a and d.



Solution:

The cut sets between a and d are

$\{ab, af\}$, $\{ab, bf, fe\}$, $\{ab, bf, ce, ed\}$, $\{af, fb, bc\}$, $\{af, fb, ce, cd\}$, $\{bc, fe\}$, $\{bc, ce, ed\}$, $\{fe, ec, cd\}$, $\{cd, ed\}$.

Capacities of these cut sets are, respectively, 9, 11, 17, 11, 10, 7, 13, 6, 10

(The capacity of the cut-set $\{ab, af\}$ is $= wt(ab) + wt(af) = 5 + 4 = 9$).

Thus, 6 is the minimum capacity, and this is possessed by the cut set $\{fe, ec, cd\}$. The possible maximum flow is therefore 6 units

Net work flows:

Consider the transport networks like the networks of telephone lines, rail roads, oil or water pipelines. To know the maximum rate of flow i.e., possible from one station to another, we consider a weighted connected graph in which

- i) vertices represent stations
- ii) edges represent the lines of the network
- iii) weight of an edge represents the capacity of the line namely, the maximum amount of flow possible per unit of time

Rate of commodity = number of telephone calls, number of trains, amount of water or oil.

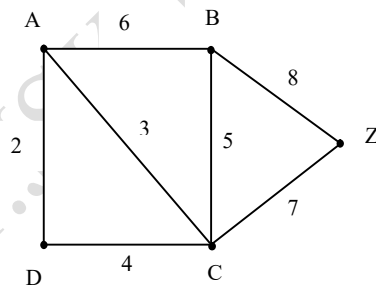
Max – flow and Min – cut Theorem:

Theorem:

The maximum possible flow between two vertices A and B in a transport network is equal to the minimum of the capacities of all cut-sets w. r. t. A and B.

Example:

For the network shown below, determine the maximum flow between A and Z by identifying a cut-set of minimum capacity



Solution:

The cut-sets w. r. t. the vertices A and Z with their capacity are

$$S_1 = \{AB, AC, AD\} = 11$$

$$S_2 = \{BZ, CZ\} = 15$$

$$S_3 = \{AB, AC, DC\} = 13$$

$$S_4 = \{BZ, BC, AC, DC\} = 20$$

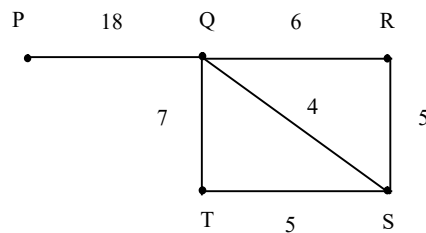
$$S_5 = \{AB, BC, CZ\} = 18$$

$$S_6 = \{AD, AC, BC, BZ\} = 18$$

Thus 11 is the minimum capacity possessed by the cut-set $\{AB, AC, AD\}$. Hence the possible maximum flow is 11 units.

Example:

Find the maximum flow possible between the vertices P and S in the network shown below



Solution:

The cut-sets w. r. t. the vertices P and S with their capacities are

$$S_1 = \{PQ\} = 18$$

$$S_2 = \{TS, QS, RS\} = 14$$

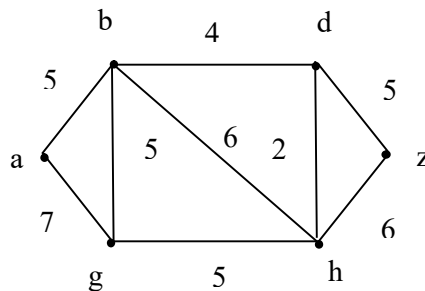
$$S_3 = \{QT, QS, RS\} = 16$$

$$S_4 = \{QR, QS, TS\} = 15$$

$$S_5 = \{QT, QS, QR\} = 17$$

Thus 14 is the minimum capacity possessed by the cut-set $\{TS, QS, RS\}$. Hence the max flow is 14 units.

Example: For the given network



Consider the two cuts

(i) cut C_1 with vertex partition $Q = \{a, b, g\}$, $Q' = \{d, h, z\}$

(ii) cut C_2 with vertex partition $Q = \{a, b, d, g, h\}$, $Q' = \{z\}$.

Find the capacity of each of these cuts.

Solution:

The cut set C_1 with vertex partition $Q = \{a, b, g\}$, $Q' = \{d, h, z\}$ is

$$C_1 = \{bd, bh, gh\}$$

$$\text{Capacity} = 4 + 6 + 5 = 15.$$

The cut C_2 with vertex partition $Q = \{a, b, d, g, h\}$, $Q' = \{z\}$ is

$$C_2 = \{dz, zh\};$$

$$\text{Capacity} = 5 + 6 = 11.$$